

Series of exercises 2: The complex numbers

Exercise 1 Prove that for all complex number $z \in \mathbb{C}$, we have

1. $|z|^2 = z \cdot \bar{z}$.
2. $-|z| \leq \operatorname{Re}(z) \leq |z|$ and $-|z| \leq \operatorname{Im}(z) \leq |z|$.

Exercise 2 Let z_1 and z_2 two complex numbers. Show that

1. $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$.
2. $|z_1 + z_2| \leq |z_1| + |z_2|$.
3. $||z_1| - |z_2|| \leq |z_1 - z_2|$.
4. $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.

Exercise 3 Solve the given equations, where $z \in \mathbb{C}$ is a complex number.

1. $2z + 6\bar{z} = 3 + 2i$.
2. $|z + i| = |z - i|$.
3. $\left| \frac{z - 12}{z - i8} \right| = \frac{5}{3}$ and $\left| \frac{z - 4}{z - 8} \right| = 1$.

Exercise 4 Let $z = \frac{-1 + i\sqrt{3}}{2} \in \mathbb{C}$ be a complex number.

1. Express z in the trigonometric and the exponential representations.
2. Compute $z \cdot \bar{z}$ and z^n where n is a natural number. Express $1 + z$ in function of $\bar{z}$.

Exercise 5 Solve the given equations, where $z \in \mathbb{C}$ is a complex number. The solutions must be given in both algebraic and trigonometric representation.

1. $z^3 + i = 0$.
2. $z^8 = \frac{1+i}{\sqrt{3}-i}$.

Exercise 6 Find the exact value of the expression $(1 - i\sqrt{3})^3 (1 + i)^{10}$.

1. Let $f(n) = \left(\frac{1+i}{\sqrt{2}}\right)^n + \left(\frac{1-i}{\sqrt{2}}\right)^n$. Determine the sum $f(1990) + f(1994)$.
2. If $z + \frac{1}{z} = 1$, then find the exact value of $z^{158} + z^{152} + \frac{2}{z^{122}}$.

Exercise 7 By using Euler's formulas, prove the identities

1. $\sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3x$.
2. $\cos^4 x = \frac{1}{8} \cos 4x + \frac{1}{2} \cos 2x + \frac{3}{8}$.
3. $\cos 5x = 16 \cos^5 x - 20 \cos^3 x + 5 \cos x$.
4. $\sin 5x = \sin x (16 \cos^4 x - 12 \cos^2 x + 1)$.